



Department of Computer Science and Engineering
Final Examination Spring 2024
CSE 321: Operating Systems

Duration: 2 Hours

Total Marks: 45

Answer the following questions.
Figures in the right margin indicate marks.

1. a) Dipu is currently developing software for a multi-threaded application where threads compete for access to shared resources. However, due to performance constraints, resources cannot be preempted. Describe a situation where circular wait arises among threads and suggest strategies to mitigate deadlock. [3]

CO4

- b) Assume that there are five processes, P1 through P5 and 4 types of resources. At T0 time we have the following system state:

	Allocation					Max					Available			
	A	B	C	D		A	B	C	D		A	B	C	D
P1	5	2	1	1		8	3	3	4		4	6	3	0
P2	5	4	3	4		10	8	6	6					
P3	4	1	5	4		9	5	5	4					
P4	2	1	1	2		2	3	4	3					
P5	2	0	2	1		6	1	3	1					

- i. Calculate the Need Matrix. Is this system in a safe state? If yes, then find the safe sequence using Banker's Safety algorithm otherwise, provide the necessary explanation. [1+3]
- ii. P1 requests for (2 1 1 0), check the validity of the request. If the request is valid, does the system enter a deadlock? [1+3]

2. a) At a particular time, the snapshot of the Main memory is given below for **dynamic** partitioning where **gray portions of the memory represent occupied spaces**. Apply **worst fit** and **first fit** algorithms to place processes with the space requirement of **P1=50K, P2=600K, P3=250K, P4=80K and P5=150K (in order)**. Explain which algorithm makes the most effective use of memory. [4+1]

300K	55K	675K	500K	230K
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- b) How does TLB help to shorten the effective memory access time? Explain briefly. [2]

- c) **Compare** the following scenarios and find the most efficient one in terms of the number of main memory lookups – [3]

- $\epsilon = 46$, Memory Access Time = 100, Effective Access Time = 164
- $\epsilon = 73$, Memory Access Time = 280, Effective Access Time = 395

- d) Zahin is managing a computer lab with several PCs, each running multiple applications concurrently. Occasionally, users complain about sluggish performance and application crashes. Upon investigation, Zahin discovers that the system is experiencing memory fragmentation due to variable partitioning in main memory. **Discuss** how memory fragmentation can occur in this scenario and its impact on system performance. **What** strategies would you implement to mitigate memory fragmentation and improve system stability? [3]

- e) []

Page no	Frame no
0	6
1	11
2	4
3	10
4	8
5	9

Given the page table for a process, Find out the logical addresses of the following physical addresses. If the address is invalid, write “Invalid”.

- 18 ((10010)_b)
- 61 ((111101)_b)
- 27 ((11011)_b)

The system you are using has **48 bytes** of main memory and the memory is of the following structure(horizontal).

0	1	2	3	4	5	6	7	8	9	10	11
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3. **CO** a) Consider a computer with a main memory that has **4 frames** and a page reference string of **1-6** pages: **[2,1,2,3,4,5,3,2,6,1,4,5,3]**. The page reference string represents the order in which the pages are accessed by a program. **Apply LRU & OPT** algorithm to simulate the page replacement that occurs when the main memory can hold at most 4 pages at a time. **Record** the number of page faults and **compare** the result. **Mention** which algorithm performs better in this scenario. **[5+1]**
- b) **Explain** why implementing OPT is challenging. **Share** an idea to overcome this challenge. **[2]**
- c) Arif Shakil is managing a server environment where multiple processes run concurrently, each requiring varying amounts of memory. However, physical memory resources are limited, leading to concerns about efficient memory management and system performance. **Discuss** how demand paging can address these challenges and improve system efficiency. **[2]**
4. **CO** a) Consider, in a disk average **seek time is 15 ms**, **disk rotation 2000 rpm**. There are **16 platters** in the disk and **one surface** is being used. Platters have **256 tracks** in each surface and **64 sectors** in each track. Each sector can store **64 kb** data. **Find** out average access time for transferring **1 GB** data. **[3]**
- b) Avijit, a novice OS developer, who is unfamiliar with Unix Inode, is tasked with creating a file storage system for a digital library application. This library hosts a wide range of resources, including e-books, academic papers, research articles, and multimedia files that may get created, modified and extended regularly. Avijit must select a disk allocation strategy that ensures direct access to files and eliminates external fragmentation. **Suggest** the most applicable solution in this scenario with proper reasoning. **[3]**
- c) **Briefly discuss** the potential drawbacks of the strategy selected above. **[2]**